

CSS & HTML Integration: Styling the Web

This presentation explores CSS and its integration with HTML. We'll cover the basics, benefits, and various methods of integration. Learn how to enhance your web design skills.

Week 04 ISTM Web Based Programming

T by Taukir Alam



What is CSS?

Cascading Style Sheets

CSS stands for Cascading Style Sheets.
It is a stylesheet language.

Separates Design from Structure

CSS separates the design from the structure of a webpage. HTML handles content, CSS handles visuals.

Controls Layout, Colors, Fonts, and More

CSS controls layout, colors, fonts, and visual aspects. It is crucial for web aesthetics.

Why Use CSS?

1 Consistency Across Pages

CSS ensures a consistent look and feel. It makes the design uniform across all pages.

2 Easier Maintenance

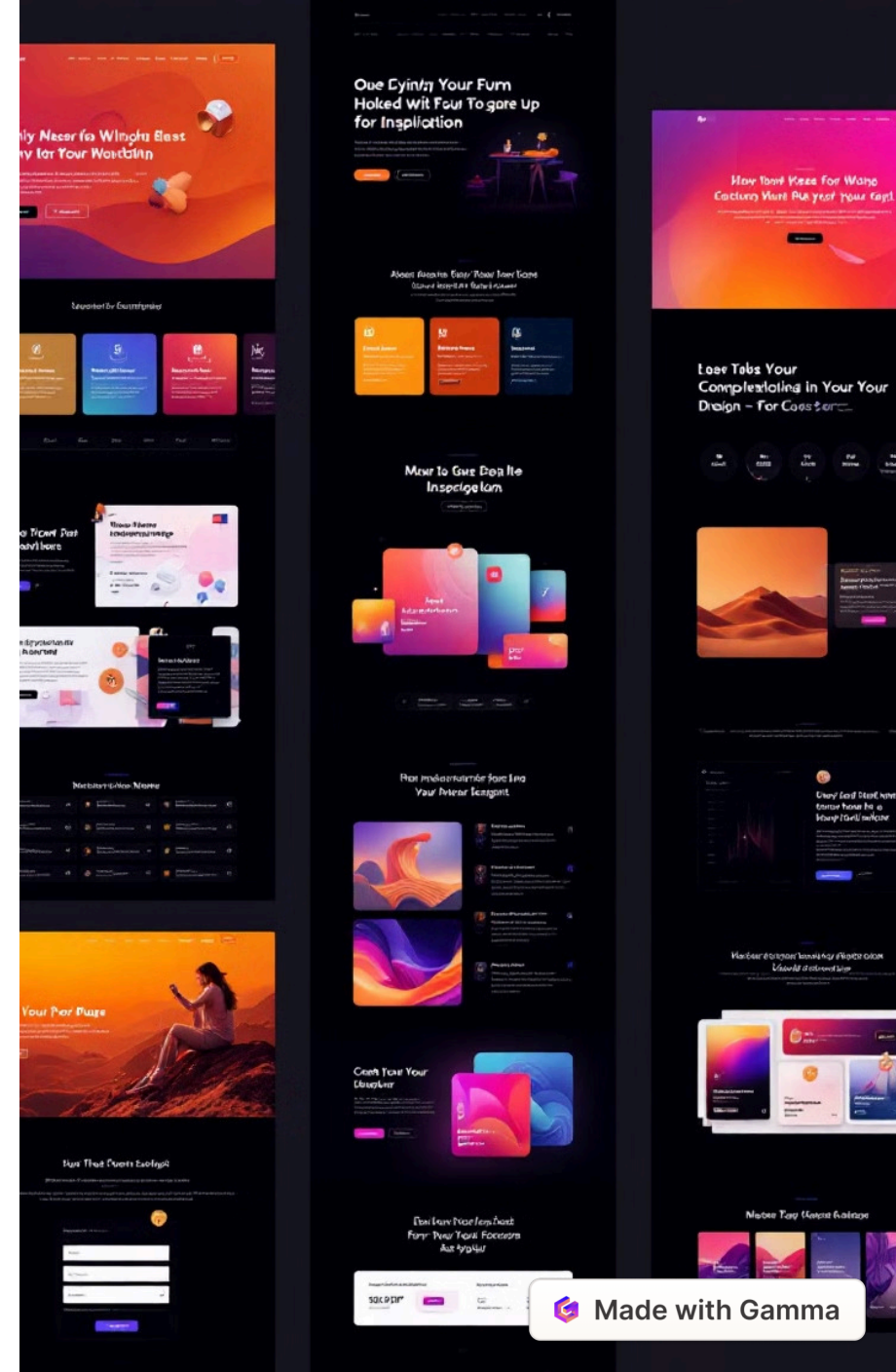
CSS simplifies website maintenance and updates. Changes can be made easily.

3 Faster Load Times

CSS promotes faster loading by reducing HTML clutter. This improves website performance.

4 Enhanced User Experience

CSS enhances the user experience. A well-styled site is more engaging.



CSS Selectors



Element Selectors

These select HTML elements directly. For example, **p** selects all paragraphs.



Class Selectors

These select elements with a specific class. Use **.class-name** to select.



ID Selectors

These select a unique element with an ID. Use **#id-name** to select.

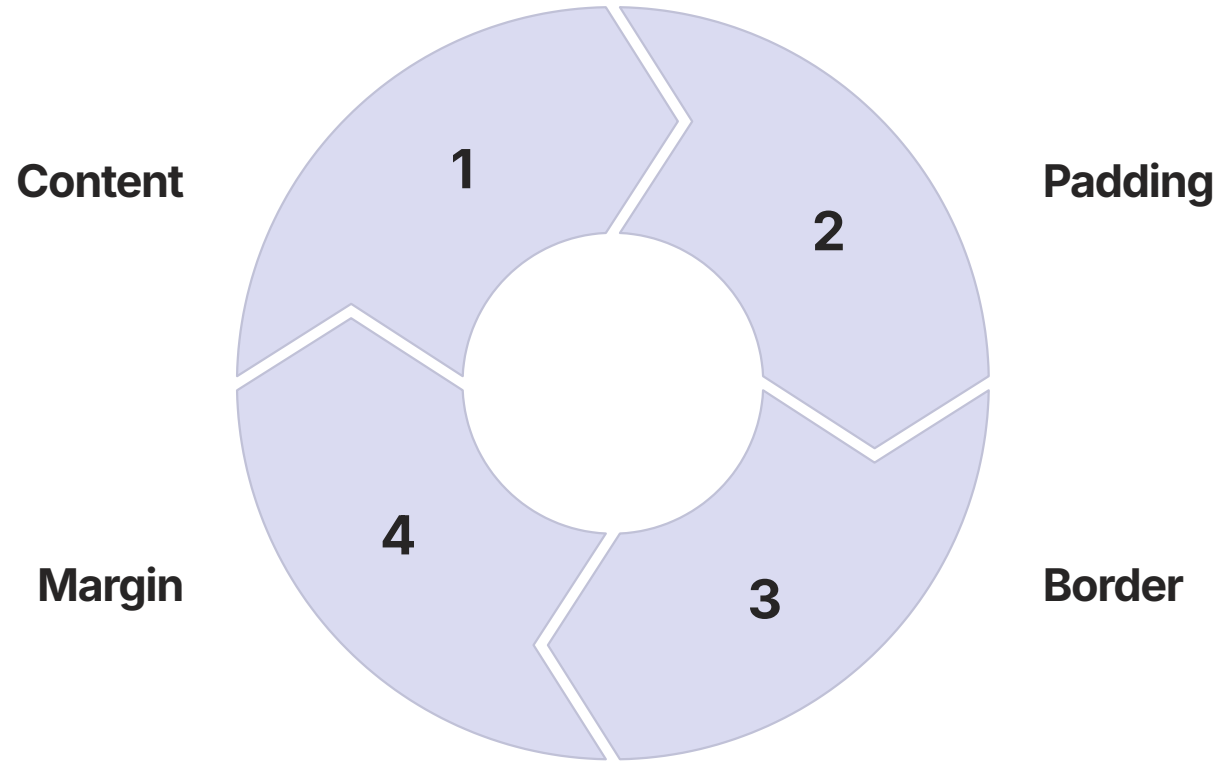


Attribute Selectors

Select elements based on their attributes. e.g., **a[href]** selects links with href.

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CSS Box Model



The CSS box model defines how HTML elements are rendered. It includes content, padding, border, and margin. Understanding the box model is key to layout control.

CSS Colors

1

Hexadecimal

Uses hex codes (e.g., **#RRGGBB**). Offers a wide range of colors.

2

RGB / RGBA

Specifies colors using red, green, and blue values. RGBA adds alpha for transparency.

3

HSL / HSLA

Hue, saturation, and lightness. HSLA adds alpha for transparency.

4

Color Names

Uses predefined color names. (e.g., **red**, **blue**).


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Integrating CSS with HTML

Inline Styles

CSS applied directly within HTML elements using the **style** attribute.

Internal Stylesheet

CSS placed inside the tag within the HTML document's **head**.

External Stylesheet

CSS defined in separate **.css** files linked to HTML using the **link** tag.

Inline CSS

1

Using the style Attribute

Apply CSS directly using the **style** attribute within HTML elements.

2

Pros and Cons

Simple but lacks maintainability. Good for small, specific styling.

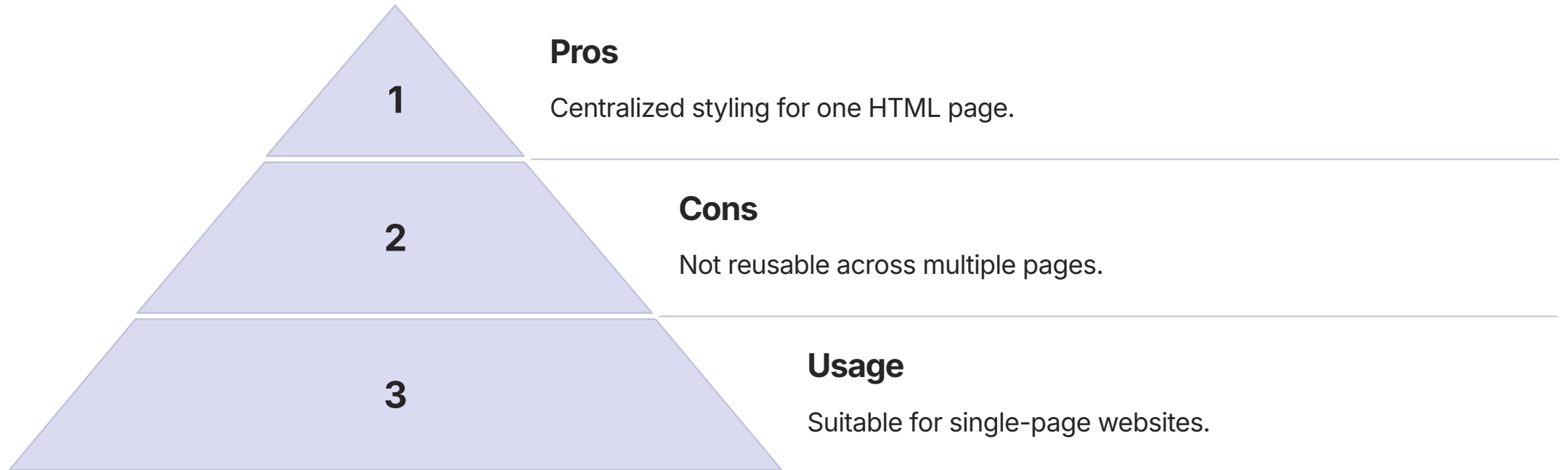
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When to Use Inline Styles

Use for quick fixes or dynamic styles via JavaScript.

`<style,`

Internal CSS

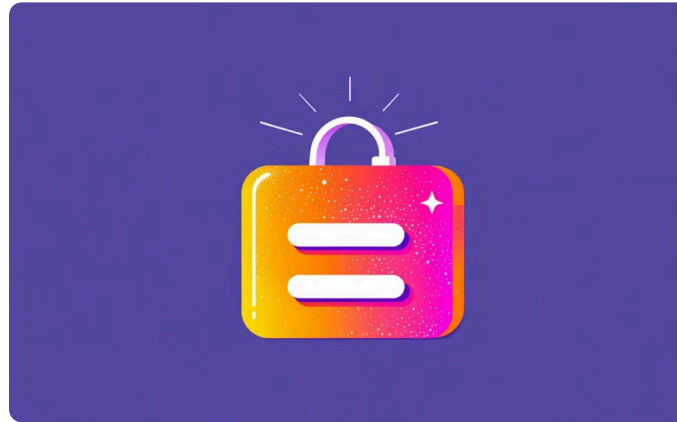


External CSS



Separate CSS File

Create a separate **.css** file for styles.



Linking to HTML

Link CSS using the **link** tag in HTML **head**.



Benefits

Reusable, maintainable, and organized.